Atlas 200I DK A2 开发者套件
23.0.RC3

AscendC 算子开发指南

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https://www.huawei.com/cn/psirt/vul-response-process
如企业客户须获取漏洞信息，请参见如下网址：
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1 环境准备

更新 CANN 软件

步骤1 切换至root用户，并进入默认下载目录/home/HwHiAiUser/Downloads。

```
su root
cd /home/HwHiAiUser/Downloads
```

步骤2 单击**链接**，获取8.0.RC1.alpha001版本的CANN开发套件包Ascend-cann-toolkit_8.0.RC1.alpha001_linux-aarch64.run，并上传至当前目录。

步骤3 增加对软件包的可执行权限。

```
chmod +x Ascend-cann-toolkit_8.0.RC1.alpha001_linux-aarch64.run
```

步骤4 卸载环境上原有的Ascend-cann-toolkit开发套件包。

```
./Ascend-cann-toolkit_8.0.RC1.alpha001_linux-aarch64.run --uninstall
```

步骤5 删除软件包所在目录下的文件。

```
rm -rf /usr/local/Ascend/ascend-toolkit/*
```

步骤6 静默安装最新版本的Ascend-cann-toolkit开发套件包，即自动签署**华为企业业务最终用户许可协议（EULA）**。

```
./Ascend-cann-toolkit_8.0.RC1.alpha001_linux-aarch64.run --full --quiet
```

步骤7 安装protobuf依赖。

```
pip install protobuf==3.20.0
```

步骤8 执行如下命令配置环境变量。

```
echo "source /usr/local/Ascend/ascend-toolkit/set_env.sh" >> /root/.bashrc
```

步骤9 执行如下命令使环境变量生效。

```
source /root/.bashrc
```

----结束

获取算子 samples 代码

在开发者套件连接个人PC并且网络连接正常的情况下，执行如下命令返回至用户根目录，在根目录下载算子samples代码。

```
cd ~
git clone https://gitee.com/ascend/samples.git
```

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2 AddCustom 算子样例执行

AddCustom算子样例支持多种方式调用执行，以下内容以ACLNN接口和核函数两种常用的方式为例进行介绍，更多方式请参考**样例仓**中的readme指导。

ACLNN 接口调用执行

步骤1 进入AddCustom算子样例目录。

```
cd ~/samples/operator/AddCustomSample/
```

目录示例如下：

```
/root/[redacted]/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom
(base) root@davinci-mini:~/[redacted]/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom# ls
CMakeLists.txt  CMakePresets.json  README.md  build.sh  cmake  framework  op_host  op_kernel  scripts
(base) root@davinci-mini:~/[redacted]/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom#
```

步骤2 执行编译打包命令生成自定义算子包。

```
cd FrameworkLaunch/AddCustom
./build.sh
```

编译成功结果示例：

```
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/ascend310p/add_custom/AddCustom_1e04ee05ab491cc5ae9c3d5c9ee8950b.o
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/ascend910/add_custom/AddCustom_1e04ee05ab491cc5ae9c3d5c9ee8950b.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/ascend910/add_custom/AddCustom_1e04ee05ab491cc5ae9c3d5c9ee8950b.o
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/ascend910b/add_custom/AddCustom_1e04ee05ab491cc5ae9c3d5c9ee8950b.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/ascend910b/add_custom/AddCustom_1e04ee05ab491cc5ae9c3d5c9ee8950b.o
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/config/ascend310b/add_custom.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/config/ascend310b/binary_info_config.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/config/ascend310p/add_custom.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/config/ascend310p/binary_info_config.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/config/ascend910/add_custom.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/config/ascend910/binary_info_config.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/config/ascend910b/add_custom.json
./packages/vendors/customize/op_impl/ai_core/tbe/kernel/config/ascend910b/binary_info_config.json
./packages/vendors/customize/op_impl/ai_core/tbe/op_tiling/lib/linux/aarch64/libcust_opmaster_rt2.0.so
./packages/vendors/customize/op_impl/ai_core/tbe/op_tiling/liboptiling.so
./packages/vendors/customize/op_proto/inc/op_proto.h
./packages/vendors/customize/op_proto/lib/linux/aarch64/libcust_opsproto_rt2.0.so
./packages/vendors/customize/version.info
./upgrade.sh
CRC:[redacted]
SHA256:[redacted]
Skipping md5sum at user request

Self-extractable archive "custom_opp_ubuntu_aarch64.run" successfully created.
Copy /root/[redacted]/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom/build_out/_CPack_Packages/Linux/External/custom_opp_ubuntu_aarch64.run/custom_opp_ubuntu_aarch64.run to /root/[redacted]samples/operator/AddCustomSample/FrameworkLaunch/AddCustom/build_out/
CPack: - package: /root/[redacted]/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom/build_out/custom_opp_ubuntu_aarch64.run.json generated.
CPack: - package: /root/[redacted]/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom/build_out/custom_opp_ubuntu_aarch64.run generated.
(base) root@davinci-mini:~/[redacted]/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom#
```

步骤3 部署自定义算子包。

```
./build_out/custom_opp_ubuntu_aarch64.run
```

部署成功结果示例：

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```
(base) root@davinci-mini:~/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom# ./build_out/custom_opp_ubuntu_aarch64.run
Verifying archive integrity...  100%   SHA256 checksums are OK. All good.
Uncompressing version:1.0  100%
[runtime] [2024-02-28 09:19:21] [INFO] copy uninstall sh success
[ops_custom]upgrade framework
tensorflow [runtime] [2024-02-28 09:19:21] [INFO] replace or merge old ops framework files .g.....
[runtime] [2024-02-28 09:19:21] copy new ops framework files ......
[ops_custom]upgrade op proto
inc lib [runtime] [2024-02-28 09:19:24] [INFO] replace or merge old ops op_proto files .g.....
[runtime] [2024-02-28 09:19:24] copy new ops op_proto files ......
[ops_custom]upgrade version.info
[runtime] [2024-02-28 09:19:24] copy new version.info files ......
[ops_custom]upgrade op impl
ai_core [runtime] [2024-02-28 09:19:26] [INFO] replace or merge old ops op_impl files .g.....
[runtime] [2024-02-28 09:19:26] copy new ops op_impl files ......
[ops_custom]upgrade op api
include lib [runtime] [2024-02-28 09:19:42] [INFO] replace or merge old ops op_api files .g.....
[runtime] [2024-02-28 09:19:42] copy new ops op_api files ......
[runtime] [2024-02-28 09:19:42] [INFO] no need to upgrade custom.proto files
SUCCESS
(base) root@davinci-mini:~/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom#
```

步骤4 通过应用程序样例调用算子的ACLNN接口进行测试。

```
cd ~/samples/operator/AddCustomSample/FrameworkLaunch/AclNNInvocation
bash run.sh
```

测试结果示例：

```
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- env INC_PATH: /usr/local/Ascend/ascend-toolkit/latest
-- env LIB_PATH: /usr/local/Ascend/ascend-toolkit/latest/aarch64-linux/lib64
-- Configuring done
-- Generating done
-- Build files have been written to: /root/samples/operator/AddCustomSample/FrameworkLaunch/AclNNInvocation/build
INFO: cmake success!
[ 20%] Building CXX object CMakeFiles/execute_add_op.dir/operator_desc.cpp.o
[ 40%] Building CXX object CMakeFiles/execute_add_op.dir/op_runner.cpp.o
[ 60%] Building CXX object CMakeFiles/execute_add_op.dir/main.cpp.o
[ 80%] Building CXX object CMakeFiles/execute_add_op.dir/common.cpp.o
[100%] Linking CXX executable /root/samples/operator/AddCustomSample/FrameworkLaunch/AclNNInvocation/output/execute_add_op
[100%] Built target execute_add_op
INFO: make success!
INFO: execute op!
[INFO]  Set device[0] success
[INFO]  Get RunMode[0] success
[INFO]  Init resource success
[INFO]  Set input success
[INFO]  Copy input[0] success
[INFO]  Copy input[1] success
[INFO]  Create stream success
[INFO]  Execute aclnnAddCustomGetWorkspaceSize success, workspace size 0
[INFO]  Execute aclnnAddCustom success
[INFO]  Synchronize stream success
[INFO]  Copy output[0] success
[INFO]  Write output success
[INFO]  Run op success
[INFO]  Reset Device success
[INFO]  Destory resource success
INFO: acl executable run success!
test pass

################################
INFO: you have passed the Precision!
################################

(base) root@davinci-mini:~/samples/operator/AddCustomSample/FrameworkLaunch/AclNNInvocation#
```

----结束

核函数调用执行

步骤1 进入AddCustom算子样例核函数调用执行目录。

```
cd ~/samples/operator/AddCustomSample/KernelLaunch/AddKernelInvocation
```

步骤2 运行kernellanch调用执行样例脚本。

```
bash run.sh -v Ascend310B1 -r npu
```

```
        vector_bool preg2 = pge_b8(PAT_ALLF);
                            ^
/usr/local/Ascend/ascend-toolkit/8.0.RC1.alpha001/aarch64-linux/ccec_compiler/lib/clang/15.0.5/include/__clang_cce_vector_intrinsics.h:221:1: note: 'pge_b8<std::integral_constant<Pat, Pat::ALLF>>' has been explicitly marked deprecated here
DEPRECATED_AFTER_V210 CCE_INTRINSIC vector_bool pge_b8(T dist) {
^
/usr/local/Ascend/ascend-toolkit/8.0.RC1.alpha001/aarch64-linux/ccec_compiler/lib/clang/15.0.5/include/__clang_cce_vector_intrinsics.h:57:46: note: expanded from macro 'DEPRECATED_AFTER_V210'
#define DEPRECATED_AFTER_V210 __attribute__((deprecated))
                                             ^
In file included from /root/libotao/samples/operator/AddCustomSample/KernelLaunch/AddKernelInvocation/add_custom.cpp:7:
In file included from /usr/local/Ascend/ascend-toolkit/latest/compiler/tikcpp/tikcfw/kernel_operator.h:27:
In file included from /usr/local/Ascend/ascend-toolkit/latest/compiler/tikcpp/tikcfw/interface/kernel_operator_intf.h:38:
In file included from /usr/local/Ascend/ascend-toolkit/latest/compiler/tikcpp/tikcfw/interface/kernel_operator_vec_cmpsel_intf.h:35:
/usr/local/Ascend/ascend-toolkit/latest/compiler/tikcpp/tikcfw/impl/dav_m300/kernel_operator_vec_cmpsel_impl.h:1017:56: warning: expected type of ops in vector loop cond is uint16_t. unexpected types may cause errors [-Wcce-compat]
        for (uint16_t i = 0; i < (uint16_t)repeatTimes / 2; ++i) {
                                                  ^
/usr/local/Ascend/ascend-toolkit/latest/compiler/tikcpp/tikcfw/impl/dav_m300/kernel_operator_vec_cmpsel_impl.h:1037:56: warning: expected type of ops in vector loop cond is uint16_t. unexpected types may cause errors [-Wcce-compat]
        for (uint16_t i = 0; i < (uint16_t)repeatTimes % 2; ++i) {
                                                  ^
75 warnings generated.
[ 66%] Building CCE object cmake/npu/CMakeFiles/add_npu.dir/__/__/main.cpp.o
[100%] Linking CCE executable ../../../add_npu
[100%] Built target add_npu
/root/samples/operator/AddCustomSample/KernelLaunch/AddKernelInvocation
INFO: compile op on ONBOARD succeed!
INFO: execute op on ONBOARD succeed!
test pass
```

----结束

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3 新建算子工程开发算子

步骤1 创建工程与IR json文件。

1. 执行如下命令返回根目录。
```
cd ~
```
2. 创建AddCustom.json文件。
```
vim AddCustom.json
```
3. 键盘按下“i”键进入INSERT模式，复制以下内容至文件，以**AddCustom**为例：
```
[
  {
    "op": "AddCustom",
    "language": "cpp",
    "input_desc": [
      {
        "name": "x",
        "param_type": "required",
        "format": [
          "ND"
        ],
        "type": [
          "float16"
        ]
      },
      {
        "name": "y",
        "param_type": "required",
        "format": [
          "ND"
        ],
        "type": [
          "float16"
        ]
      }
    ],
    "output_desc": [
      {
        "name": "z",
        "param_type": "required",
        "format": [
          "ND"
        ],
        "type": [
          "float16"
        ]
      }
    ]
  }
]
```

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4. 编辑完成后，按“ESC”键退出INSERT模式，执行**:wq**保存AddCustom.json文件。

步骤2 使用msopgen工具创建Ascend C算子工程。

```
/usr/local/Ascend/ascend-toolkit/latest/python/site-packages/bin/msopgen gen -i AddCustom.json -f tf -c ai_core-ascend310b -lan cpp -out ~/AddCustom
```

```
(base) root@davinci-mini:~# /usr/local/Ascend/ascend-toolkit/latest/python/site-packages/bin/msopgen gen -i AddCustom.json -f tf -c ai_core-ascend310b -lan cpp -out AddCustom
2024-02-29 17:45:19 (1126882) - [INFO] Start to generate AI Core operator files.
2024-02-29 17:45:19 (1126882) - [INFO] Start to parse the ir template:/root/AddCustom.json
2024-02-29 17:45:19 (1126882) - [INFO] Start to parse the op: AddCustom
2024-02-29 17:45:19 (1126882) - [INFO] Start to parse the input_desc: x
2024-02-29 17:45:19 (1126882) - [INFO] Start to parse the input_desc: y
2024-02-29 17:45:19 (1126882) - [INFO] Start to parse the output_desc: z
2024-02-29 17:45:19 (1126882) - [WARNING] The "attr" value is invalid or no "attr" exists in the map.
2024-02-29 17:45:19 (1126882) - [INFO] Start to check the type and format between the inputs/outputs in IR template.
2024-02-29 17:45:19 (1126882) - [INFO] Start to generate a new project.
2024-02-29 17:45:19 (1126882) - [INFO] File /root/AddCustom/cmake/config.cmake generated successfully.
2024-02-29 17:45:19 (1126882) - [INFO] File /root/AddCustom/op_host/add_custom_tiling.h generated successfully.
2024-02-29 17:45:19 (1126882) - [INFO] File /root/AddCustom/op_host/add_custom.cpp generated successfully.
2024-02-29 17:45:19 (1126882) - [INFO] File /root/AddCustom/op_kernel/add_custom.cpp generated successfully.
2024-02-29 17:45:19 (1126882) - [INFO] File /root/AddCustom/framework/tf_plugin/tensorflow_add_custom_plugin.cc generated successfully.
2024-02-29 17:45:19 (1126882) - [INFO] File /root/AddCustom/framework/tf_plugin/CMakeLists.txt generated successfully.
2024-02-29 17:45:19 (1126882) - [INFO] Generation completed.
(base) root@davinci-mini:~#
```

步骤3 开发算子的host侧代码及kernel侧代码。

可参考Samples仓AddCustom算子样例，host侧：

```
cp ~/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom/op_host/* ~/AddCustom/op_host/
```

kernel侧：

```
cp ~/samples/operator/AddCustomSample/FrameworkLaunch/AddCustom/op_kernel/* ~/AddCustom/op_kernel/
```

步骤4 修改算子工程中配置文件CMakePresets.json，将CMakePresets.json中ASCEND_CANN_PACKAGE_PATH的value值修改为实际CANN包安装目录/usr/local/Ascend/ascend-toolkit/latest。

1. 执行如下命令打开配置文件。
   ```
   cd ~/AddCustom
   vim CMakePresets.json
   ```
2. 使用键盘的方向键令光标移动到ASCEND_CANN_PACKAGE_PATH的值位置，按下键盘上“i”键进入到INSERT模式，修改为CANN包安装目录

```
        "ASCEND_COMPUTE_UNIT": {
            "type": "STRING",
            "value": "ascend310b"
        },
        "ENABLE_TEST": {
            "type": "BOOL",
            "value": "True"
        },
        "vendor_name": {
            "type": "STRING",
            "value": "customize"
        },
        "ASCEND_CANN_PACKAGE_PATH": {
            "type": "PATH",
            "value": "/usr/local/Ascend/ascend-toolkit/latest"
        },
        "ASCEND_PYTHON_EXECUTABLE": {
            "type": "STRING",
            "value": "python3"
        },
```

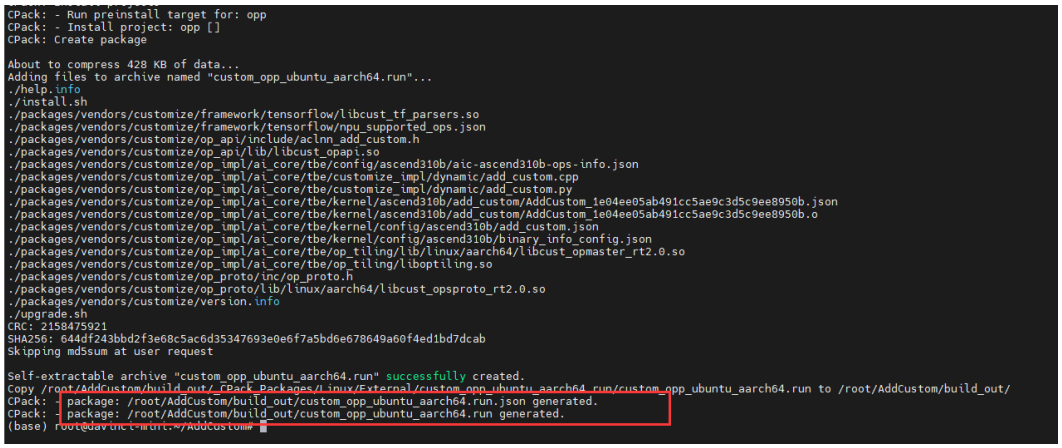
修改完成后按“ESC”键退出INSERT模式，执行**:wq**保存文件。

步骤5 编译生成算子安装包。

在AddCustom目录下执行如下命令编译自定义AddCustom算子，生成算子安装包。

```
./build.sh
```

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